Lesson 6: If-Elif-Else — Many Choices 🔀🔀🔀

> **Goal:** Use `elif` to handle multiple conditions cleanly, and understand nested `if` statements.

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1. The Problem with Many Choices

In Lesson 5, you learned `if/else` — two choices. But what if you have **more than two** choices?

Imagine a traffic light: it can be red, yellow, or green. With only `if/else`, you would have to nest many levels deep:

```
# Lesson 5 style — gets messy quickly!
if color == "red":
    print("Stop")
else:
    if color == "yellow":
        print("Slow down")
    else:
        print("Go")
```

Python gives us a cleaner tool: `elif`.

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2. `elif` — Else If

`elif` means "else if" — check another condition if the previous one was False.

```
color = input("Traffic light color: ")

if color == "red":
    print("Stop")
elif color == "yellow":
    print("Slow down")
else:
    print("Go")
```

How it works:

1. Check `color == "red"`. If True → print "Stop", skip the rest.
2. If False → check `color == "yellow"`. If True → print "Slow down", skip the rest.

3. If False → run `else` → print "Go".

Only ONE branch runs — the first condition that is True.

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3. The Structure

```
if condition1:
    code for condition1
elif condition2:
    code for condition2
elif condition3:
    code for condition3
else:
    code if none of the above
```

- You can have **as many `elif` as you need**.
- The `else` at the end is optional — it catches everything that didn't match.
- Python checks conditions **from top to bottom** and stops at the first True one.

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4. Grade Example

```
score = int(input("Enter your score: "))

if score >= 90:
    print("A")
elif score >= 80:
    print("B")
elif score >= 70:
    print("C")
elif score >= 60:
    print("D")
else:
    print("F")
```

Example runs:

Input	Output	Why
95	A	95 ≥ 90 → first condition True
83	B	83 < 90, but 83 ≥ 80 → second condition True
55	F	None of the conditions matched → else

> 💡 Notice: we don't need to write `score >= 80 and score < 90` for B. Because if we reach the `elif score >= 80` check, we already know `score < 90` (otherwise the first `if` would have caught it).

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5. Only One Branch Runs

This is very important! Even if multiple conditions are True, **only the first True one runs**.

```
x = 15

if x > 10:
    print("greater than 10")   # This runs
elif x > 5:
    print("greater than 5")    # This is SKIPPED (even though 15 > 5 is True)
else:
    print("5 or less")         # This is SKIPPED
```

Output:

```
greater than 10
```

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6. Nested `if`

You can put an `if` statement **inside** another `if` statement. This is called **nested if**.

```
age = int(input("Enter age: "))
has_ticket = input("Do you have a ticket? (yes/no): ")

if age >= 18:
    if has_ticket == "yes":
        print("Welcome! Enjoy the show.")
    else:
        print("Please buy a ticket first.")
else:
    print("Sorry, this show is for adults only.")
```

The inner `if` only runs if the outer `if` is True.

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7. Season Example

```
month = int(input("Enter month (1-12): "))

if month == 3 or month == 4 or month == 5:
    print("Spring")
elif month == 6 or month == 7 or month == 8:
    print("Summer")
elif month == 9 or month == 10 or month == 11:
    print("Autumn")
else:
    print("Winter")
```

(We will learn `and` / `or` properly in Lesson 7, but you can see how they work here.)

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8. Menu Example

```
print("=== Menu ===")
print("1. Add")
print("2. Subtract")
print("3. Multiply")

choice = int(input("Choose (1-3): "))
a = float(input("Enter a: "))
b = float(input("Enter b: "))

if choice == 1:
    print("Result:", a + b)
elif choice == 2:
    print("Result:", a - b)
elif choice == 3:
    print("Result:", a * b)
else:
    print("Invalid choice!")
```

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9. Common Mistakes

```
# Mistake 1: elif without if
elif x > 5:     # ❌ SyntaxError! elif must come after if
    print("yes")

# Mistake 2: else with a condition
else x > 5:     # ❌ SyntaxError! else never has a condition
    print("yes")
```

```
# Mistake 3: Wrong order (conditions overlap)
score = 85
if score >= 60:
    print("D")    # ❌ This prints "D" for score 85!
elif score >= 80:
    print("B")    # This never runs for 85 because the first if already caught it

# Fix: put more specific conditions first
if score >= 80:
    print("B")    # ✅ Checks 80+ first
elif score >= 60:
    print("D")
```

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10. Vocabulary

English	Vietnamese	Meaning
elif	khác nếu	"else if" — check another condition
chain	chuỗi	Multiple conditions linked together
nested	lồng nhau	An if inside another if
branch	nhánh	One path of a multi-way decision
condition	điều kiện	An expression that is True or False
multiple	nhiều	More than two choices
fall-through	rơi xuống	When no condition matches, goes to else
default	mặc định	The else case — what happens if nothing matches

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11. Summary

✅ `elif` lets you check multiple conditions in sequence. ✅ Only the **first True** branch runs — the rest are skipped. ✅ `else` at the end catches everything that didn't match any condition. ✅ Put **more specific** conditions first, **more general** ones later. ✅ Nested `if` = an `if` inside another `if`. ✅ `elif` is cleaner than deeply nested `if/else`.

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12. Homework

🟢 **Easy**

Exercise E1. Traffic Light

Read a traffic light color (`red`, `yellow`, or `green`). Print the instruction.

- **Input:** One word — the color.
- **Output:** The instruction.

Example:

Input	Output
red	Stop
yellow	Slow down
green	Go

Exercise E2. Day Type

Read a day number (1 = Monday, 7 = Sunday). Print `weekday` for days 1–5, `weekend` for days 6–7.

- **Input:** One integer (1–7).
- **Output:** `weekday` or `weekend`.

Example:

Input	Output
3	weekday
6	weekend
7	weekend

Exercise E3. Number Sign

Read a number. Print `positive`, `negative`, or `zero`.

- **Input:** One integer.
- **Output:** `positive`, `negative`, or `zero`.

Example:

Input	Output
5	positive
-3	negative
0	zero

Exercise E4. Small, Medium, Large

Read a number. Print `small` if less than 10, `medium` if 10–99, `large` if 100 or more.

- **Input:** One integer.
- **Output:** `small`, `medium`, or `large`.

Example:

Input	Output
5	small
50	medium
200	large

Exercise E5. Vowel or Consonant (simple)

Read a single lowercase letter. Print `vowel` if it is `a`, `e`, `i`, `o`, or `u`, otherwise print `consonant`.

- **Input:** One lowercase letter.
- **Output:** `vowel` or `consonant`.

Example:

Input	Output
a	vowel
b	consonant
i	vowel

Exercise E6. Score Message

Read a score. Print `excellent` if ≥ 90, `good` if ≥ 70, `ok` if ≥ 50, `poor` if below 50.

- **Input:** One integer (0–100).
- **Output:** `excellent`, `good`, `ok`, or `poor`.

Example:

Input	Output
95	excellent
75	good
55	ok
30	poor

Exercise E7. Age Group

Read an age. Print `baby` (0–2), `child` (3–12), `teen` (13–17), `adult` (18–64), `senior` (65+).

- **Input:** One integer (0–120).
- **Output:** The age group.

Example:

Input	Output
1	baby
8	child
15	teen
30	adult
70	senior

🟡 Medium

Exercise M1. Grade with Remarks

Read a score. Print the grade (A/B/C/D/F) AND a remark on the next line.

- **Input:** One integer (0–100).

- **Output:** Two lines — grade, then remark.

Remarks: A → "Outstanding!", B → "Well done!", C → "Good effort.", D → "Keep trying.", F → "Please study more."

Example:

Input	Output
92	A
	Outstanding!
75	C
	Good effort.

Exercise M2. Month Name

Read a month number (1–12). Print the month name.

- **Input:** One integer (1–12).
- **Output:** The month name.

Example:

Input	Output
1	January
7	July
12	December

Exercise M3. Shipping Zone

Read a distance in km. Print the shipping zone: Zone A (≤ 10 km), Zone B (≤ 50 km), Zone C (≤ 200 km), Zone D (> 200 km).

- **Input:** One integer — distance in km.
- **Output:** The zone.

Example:

Input	Output
5	Zone A
30	Zone B
100	Zone C
500	Zone D

Exercise M4. Rock Paper Scissors (one player)

Read a choice (`rock`, `paper`, or `scissors`). Print the computer's choice (always `rock`) and who wins.

- **Input:** One word.
- **Output:** Two lines — computer's choice, then result.

Example:

Input	Output
rock	Computer: rock
	Draw!
scissors	Computer: rock
	Computer wins!
paper	Computer: rock
	You win!

Exercise M5. Electricity Bill

Read the number of kWh used. Calculate the bill:

- First 50 kWh: 1,500 dong/kWh
- Next 50 kWh (51–100): 2,000 dong/kWh
- Above 100 kWh: 2,500 dong/kWh
- **Input:** One integer — kWh used.
- **Output:** Total bill in dong.

Example:

Input	Output
30	45000
80	135000
120	225000

Exercise M6. Quadrant

Read two integers x and y (neither is 0). Print which quadrant the point (x, y) is in.

- **Input:** Two integers — x, then y.
- **Output:** `Quadrant 1`, `Quadrant 2`, `Quadrant 3`, or `Quadrant 4`.

(Quadrant 1: x>0, y>0. Quadrant 2: x<0, y>0. Quadrant 3: x<0, y<0. Quadrant 4: x>0, y<0.)

Example:

Input	Output
3 / 5	Quadrant 1
-2 / 4	Quadrant 2
-1 / -3	Quadrant 3
5 / -2	Quadrant 4

Exercise M7. Ticket Type

A theme park has three ticket types: child (under 6) is free, child (6–12) costs 50,000, teen (13–17) costs 70,000, adult (18+) costs 100,000.

- **Input:** One integer — age.
- **Output:** The ticket price.

Example:

Input	Output
4	0
8	50000

Input	Output
15	70000
25	100000

🔴 Hard

Exercise H1. Grade with Elif

A school uses this grading scale: A (90–100), B (80–89), C (70–79), D (60–69), F (0–59). Write a program using `elif` to print the correct grade.

- **Input:** One integer — the score (0 ≤ score ≤ 100).
- **Output:** One letter — the grade.

Example:

Input	Output
95	A
83	B
72	C
65	D
45	F

Exercise H2. Season

The four seasons in the Northern Hemisphere are: Spring (March–May), Summer (June–August), Autumn (September–November), Winter (December, January, February).

Write a program that reads a month number and prints the season.

- **Input:** One integer — the month (1 ≤ month ≤ 12).
- **Output:** The season name.

Example:

Input	Output
4	Spring
7	Summer
10	Autumn
1	Winter
12	Winter

Exercise H3. Triangle Type

Given three side lengths, a triangle can be:

- **Equilateral**: all three sides are equal.
- **Isosceles**: exactly two sides are equal.
- **Scalene**: all three sides are different.

Write a program that reads three side lengths and prints the triangle type.

- **Input:** Three integers on separate lines — A, B, C ($1 \le A, B, C \le 1000$).
- **Output:** `equilateral`, `isosceles`, or `scalene`.

Example:

Input	Output
5 / 5 / 5	equilateral
3 / 3 / 5	isosceles
3 / 4 / 5	scalene

Exercise H4. BMI Category

A doctor uses BMI to classify patients. BMI = weight (kg) ÷ (height (m))².

Categories:

- Underweight: $BMI < 18.5$
- Normal: $18.5 \le BMI < 25$
- Overweight: $25 \le BMI < 30$
- Obese: $BMI \ge 30$

Write a program that reads weight and height, then prints the BMI category.

- **Input:** Two lines — weight in kg (float), then height in meters (float).
- **Output:** The BMI category.

Example:

Input	Output
40 / 1.60	Underweight
65 / 1.70	Normal
80 / 1.65	Overweight
100 / 1.70	Obese

Exercise H5. FizzBuzz Single

FizzBuzz is a classic programming puzzle. For a single number:

- If divisible by both 3 and 5 → print `FizzBuzz`
- If divisible by 3 only → print `Fizz`
- If divisible by 5 only → print `Buzz`
- Otherwise → print the number itself

Write a program that reads one number and prints the correct output.

- **Input:** One integer.
- **Output:** `FizzBuzz`, `Fizz`, `Buzz`, or the number.

Example:

Input	Output
15	FizzBuzz
9	Fizz
10	Buzz
7	7

Exercise H6. Shipping Cost

A delivery company charges based on package weight:

- Up to 1 kg: 15,000 dong
- Up to 5 kg: 25,000 dong
- Up to 10 kg: 40,000 dong
- Over 10 kg: 60,000 dong

Write a program that reads the weight and prints the shipping cost.

- **Input:** One float — weight in kg.
- **Output:** The shipping cost in dong.

Example:

Input	Output
0.5	15000
3	25000
8	40000
15	60000

Exercise H7. Calculator with Operator

Write a calculator that reads two numbers and an operator, then prints the result. Handle division by zero by printing `Error: division by zero`.

- **Input:** Three lines — first number (float), operator (`+`, `-`, `*`, `/`), second number (float).
- **Output:** The result, or an error message.

Example:

Input	Output
10 / + / 5	15.0
10 / - / 3	7.0
6 / * / 7	42.0
10 / / / 2	5.0
10 / / / 0	Error: division by zero